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Decyl esters production from soybean-based oils catalyzed by lipase immobilized on differently functionalized rice husk silica and their characterization as potential biolubricants

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ABSTRACT

This study aimed the enzymatic decyl esters production by hydroesterification, a two-step process consisting of hydrolysis of refined soybean (RSBO) or used soybean cooking (USCO) oils to produce free fatty acids (FFA) and further esterification of purified FFA. Using free lipase from *Candida rugosa* (CRL), about 98% hydrolyses for both oils have been observed after 180 min of reaction using a CRL loading of 50 U g^{-1} of reaction mixture, 40 °C, and a mechanical stirring of 1500 rpm. FFA esterification with decanol in solvent-free systems was performed using lipase from *Thermomyces lanuginosus* (TLL) immobilized by physical adsorption on silica particles extracted from rice husk, an agricultural waste. For such purpose, non-functionalized (SiO_2) or functionalized rice husk silica bearing octyl (Octyl–SiO_2) or phenyl (Phe–SiO_2) groups have been used as immobilization supports. Protein amounts between 22 and 28 mg g^{-1} of support were observed. When used in the esterification, they enabled a FFA conversion of 81.3–87.6% after 90–300 min of reaction. Lipozyme TL IM, a commercial immobilized TLL, exhibited similar performance compared to TLL–Octyl–SiO_2 (FFA conversion ≈90% after 90–120 min of reaction). However, high operational stability after fifteen successive esterification batches was observed only for TLL immobilized on Octyl–SiO_2 (activity retention of ≈90% using both FFA sources). The produced decyl esters presented good characteristics as potential biolubricants according to standard methods (ASTM) and thermal analysis.

1. Introduction

Lubricants are important compounds typically used to reduce friction stress between surfaces [1]. They are used as anti-wear, antioxidants, antifoaming, demulsifying and emulsifying agents, corrosion inhibitors, hydraulic brake, and gear box fluids. In machinery, they act as engine, compressor, gear and piston oils and may be also used in soap and paint industries [2,3]. They are predominantly composed of a complex mixture of several hydrocarbons such as paraffin waxes, olefin, and polycyclic aromatics [3]. Over the last hundred years, mineral-based lubricants have established in most of the industrial and automotive applications. Accidental spoilage of these compounds (or their discarding after their use) is harmful to the environment due to their high toxicity [1,3].

In order to overcome such limitations, bio-based lubricants (or biolubricants) have gained increased acceptance due to their sustainable and eco-friendly features [3,4]. They have a much lower toxicity than mineral products, and they are produced from renewable sources. Indeed, the potential for biolubricants produced using triacylglycerols (TAGs) such as vegetable/microbial oils and animal fats or their derivatives such as waste cooking oils, greases and FFA as sustainable feedstocks to produce biolubricants to replace conventional lubricants from mineral oils has been currently viewed in literature as the most promising approach [1,3–5]. This is due to their better physicochemical

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properties than the conventional mineral lubricants such as high anti-corrosion properties, lubricity, viscosity index and flash point, low aquatic toxicity and biodegradability [1,3]. Moreover, their use could reduce possible dependency on imported mineral-based lubricants [1,4]. Several vegetable oils including soybean, palm, Jatropha, castor bean, corn, coconut, canola, rapeseed, olive and sunflower oils have been explored for the bio-based lubricants production [1,3,5]. Soybean oil is the second most widely consumed in the world – 59.48 million metric tons in 2020/2021 [6], and its use as potential feedstock for biolubricants production can be highly promising in many countries. Its use in food processing generates a large volume of used cooking oil every year, which is discharged into the natural environment. This is performed without proper treatment in many instances, thus causing negative impacts on human life and ecosystems [7,8], or may be properly processed to be eliminated, with the over-costs that this produces. Therefore, its use as feedstock in biolubricants production can be environmentally interesting (from a double point of view, use of a contaminant by-product and production of a green lubricant) and economically feasible [9,10].

Although vegetable oils have some lubricant properties, their use is limited due to the chemical instability of reactive β-hydrogen atoms in glycerol moiety that results in low oxidative and thermal stabilities [1]. Chemical modification of their structures by several techniques such as esterification, transesterification, epoxidation, hydrogenation or hydroesterification have been successfully proposed to overcome such limitations [1,4,5]. The direct transesterification of vegetable oils with mono- or polyols to produce esters with lubricant properties is scarcely described in the literature, since it requires high temperatures or long-reaction time to attain high yields due to the high viscosity in the reaction mixture that increases mass transfer limitations and can complicate the recovering of the biocatalyst [11,12]. In this context, several biolubricants has been produced via two-steps processes to minimize such problems such as: *(i)* sequential transesterification of vegetable oils with short-chain alcohols (methanol or ethanol) to produce monoalkyl esters (biodiesel) followed by transesterification of the produced monoalkyl esters with polyols using chemical catalysts [13–19], or *(ii)* hydroesterification of vegetable oils using a variety of lipases (free or immobilized) as catalysts [10,20–24]. This latter process consists of a TAG hydrolysis step to produce FFA, their purification and their subsequent esterification with proper alcohols – linear or branched mono-/polyols [25]. The hydrolysis step has been preferentially performed using non-specific lipases from dormant castor bean seeds (*Ricinus communis*) [20,26–28], or microbial ones like the those from *Candida rugosa* (CRL) [10,21–24] to fully convert TAG into FFA and glycerol. The use of these enzyme extracts, containing several lipases with different specificities, has some advantages in this process where enzyme specificity can reduce the yields [29]. The esterification of the produced FFA after eliminating the glycerin with several alcohols has been performed using heterogeneous acid catalysts or immobilized lipases [20,21,23,28,30,31]. The use of immobilized lipases has currently emerged as catalysts of these processes due to their high selectivity and mild reaction conditions that prevent the formation of by-products [10,20–22,28,32–34].

The immobilization of lipases has been performed via hydrophobic or ionic interactions, covalent attachment, entrapment or cross-linking [35–37]. However, the most popular protocol used in lipases immobilization is the physical adsorption at low ionic strength using a variety of hydrophobic supports [38,39]. This protocol is based on their peculiar catalytic mechanism known as “interfacial activation” [40–42], that permits their immobilization in monomeric form and “open conformation” on a hydrophobic support surface [39]. This protocol is simple and promotes simultaneous purification, immobilization, hyperactivation and stabilization of lipases from several sources [38,39,43,44]. Here, we propose the use of chemically modified rice husk silica (RHS) particles (to increase the support surface hydrophobicity) as support to immobilize lipases. This material has been used to immobilize enzymes in some instances with promising results [45–47]. This way, another agricultural residue product is utilized and that way reduces the ecological impact of the process and increases its economic feasibility.

The esterification is a thermodynamically controlled process where water is a reaction by-product [48,49]. Although water may be captured outside the biocatalyst [50], when the support of the biocatalyst is hydrophilic, water may become accumulate inside the biocatalysts particle [51]. The accumulation of water molecules inside the biocatalyst particle produces the enzyme inactivation or inhibition [52,53]. Moreover, the creation of this hydrophilic microenvironment can drastically reduce the partition of hydrophobic substrates to the active centers of the enzyme and promote the desorption of some lipase molecules from the support [54,55], thus decreasing their catalytic activity and stability. The use of hydrophobic supports [53,56,57] or ultrasounds [58,59] prevents the hydrophilic layer formation inside the particle.

The aim of this work was the enzymatic production of esters with biolubricant properties via a two-step process (hydroesterification) using refined (RSBO) and used soybean cooking oil (USCO) as oleaginous feedstocks. An illustrative scheme of this two-step process is shown in Fig. 1. For this, the hydrolysis reaction was conducted using a non-specific lipase (CRL) for complete conversion of both oils into FFA and glycerol [21,22,24]. In this step, an attempt to give fundamental understanding of the impact of relevant factors such as lipase amount, reaction temperature and stirring speed on the FFA production by hydrolysis reaction by a one-factor-at-a-time approach was performed. The hydrolysis step was performed in the absence of emulsifier or buffer agents that requires an intense mechanical stirring to prepare a stable water/oil interface to improve the catalytic performance of CRL [26,27]. Under such conditions, the use of free lipase may be an interesting option due to mechanical shear of heterogeneous biocatalysts (immobilized lipases) that could difficult their separation from the reaction mixture and their reusability [60]. Next, enzymatic esterification reaction of the produced and purified FFA with decanol was conducted in an ecofriendly process (solvent-free system) [61]. In this step, lipase from *Thermomyces lanuginosus* – Lipolase® 100 L (TLL) was selected as it is one of the most utilized lipases in literature [62]. The enzyme was immobilized via hydrophobic interactions (interfacial activation) on functionalized RHS (Octyl–SiO_2 and Phe–SiO_2), and they were used as heterogeneous biocatalysts. Their performance were compared with those of TLL immobilized on non-chemically modified SiO_2 particles and soluble TLL (free lipase). To show the advantages of the new heterogeneous biocatalyst, we have also compared its performance with that of the most popular commercial immobilized formulation of this enzyme – Lipozyme TL IM. This is a commercial heterogeneous biocatalyst supplied by Novozymes S.A. prepared via physical adsorption of TLL on silica gel particles [62]. This commercial biocatalyst has been used in the interesterification of triacylglycerols due to its high selectivity degree [63]. Both hydrophobic supports (Octyl–SiO_2 and Phe–SiO_2) were chosen to perform this study because provide biocatalysts with high catalytic activity and reusability in esters production [46,47,64]. The application of these biocatalysts (non- and commercial immobilized TLL) in decyl esters production using soybean oils as feedstcoks has not been reported in the literature yet. In this pioneer study, the physicochemical characteristics of the produced decyl esters were determined according to standard methods. The thermal analysis of the produced esters was also performed by TGA/DSC analysis.

2. Materials and methods

2.1. Materials

TLL (Lipolase® 100 L), a soluble lipase with hydrolytic activity of 27300 U mL^{-1} of crude extract and protein concentration of 17.2 mg mL^{-1}, and CRL – type VII, a powder lipase with hydrolytic activity of 16200 U g^{-1} of crude extract and protein concentration of 24.5 mg g^{-1}, were acquired from Sigma-Aldrich (St. Louis, MO, USA).

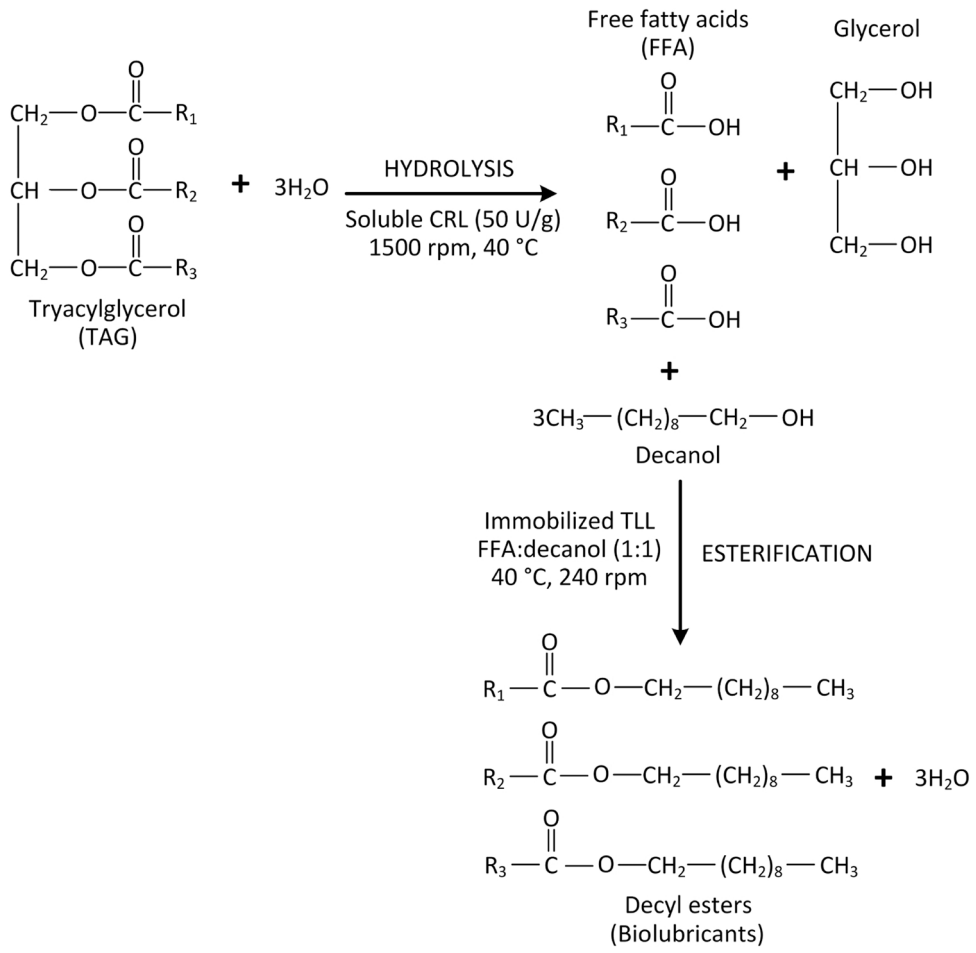


Fig. 1. Illustrative scheme of decyl esters production via enzymatic hydroesterification of soybean-based oils (refined and used soybean cooking oils).

Lipozyme TL IM, a commercial biocatalyst with a immobilized protein concentration of 57 mg g^{-1} of support [65,66], was acquired from Novozymes S.A. (Araucária, PR, Brazil). Rice husks were donated by Arroz Rei Ouro Ltda. (Itajubá, MG, Brazil). Triethoxy(octyl)silane (OCTES), triethoxy(phenyl)silane (PheTES) and decanol were also acquired from Sigma-Aldrich. RSBO (Liza) was purchased at a local market (Alfenas, MG, Brazil). Its fatty acid composition has been reported in a previous study [27]. USCO was obtained after being once used for French fries production at a fast-food restaurant within Alfenas, MG, Brazil. The samples were prepared and stored as previously reported [67]. All other chemicals were of analytical grade from Synth® (São Paulo, SP, Brazil).

2.2. Enzymatic hydrolysis of RSBO

The enzymatic hydrolysis of RSBO to produce FFA was conducted in a 250 mL closed stirred-tank reactor containing a fixed mass of reaction mixture (50 g) prepared using a ratio oil:water of 50% (m m^{-1}) immersed in a thermostatic water bath shaker. The continuous mechanical stirring was provided by an overhead motor stirrer having one steel helical impeller. The reaction was started by adding a suitable amount of crude lipase extract from CRL. Samples (1 g) were periodically withdrawn from the reactor and transferred to a 125 mL conical flask, followed by adding 10 mL of 95% v v^{-1} ethanol. Phenolphthalein was added to this mixture and titrated with NaOH solution (30 mmol L^{-1}). The reactions were performed with two replications and the experimental data are represented as mean ± error. The hydrolysis percentage (HP – %) was determined as follows (Eq. (1)) [68].

$$HP = \frac{V_{NaOH} \times 10^{-3} \times M_{NaOH} \times MM_{FFA}}{m_{sample} \times f_{SBO}} \times 100 \quad (1)$$

where: V_{NaOH} is the volume of NaOH solution consumed during titration (mL), M_{NaOH} is the NaOH concentration (30 mmol L^{-1}), MM_{FFA} is average molecular mass of FFA from soybean oils – RSBO or USCO (278.6 g mol^{-1}), m_{sample} is the mass of the sample taken (1 g), and f_{SBO} is the mass fraction of oils in the reaction mixture.

Different amounts of enzyme from 30 U g^{-1} (or 1.85 mg of powder lipase preparation g^{-1} of reaction mixture) to 60 U g^{-1} (or 3.7 mg of crude lipase extract g^{-1} of reaction mixture) were used. The temperature range was 25 ± 1–50 ± 1 °C. The stirring speed was varied between 0 and 2000 rpm.

2.2.1. Enzymatic hydrolysis of USCO

The FFA production using USCO as the starting material was conducted as aforementioned in Section 2.2. The experimental conditions were those established for RSBO hydrolysis: CRL loading of 50 U g^{-1} (or 3.1 mg of crude lipase extract per gram of reaction mixture, 40 °C and mechanical stirring speed of 1500 rpm). Hydrolysis reactions were performed using a ratio oil:water of 50% and 40% (m m^{-1}).

2.3. FFA extraction

The reaction mixture (50 g) was added to a separation funnel and 50 mL distilled water at 50 °C (volume ratio of 1:1) was added. The upper phase (consisting of FFAs) was separated from the lower phase (aqueous phase). Water washing procedure was performed four times in

order to purify the FFAs. Subsequently, anhydrous sodium sulfate (20% m m^{-1}) was added to the purified FFA for 18 h at 25 °C under static conditions in order to remove any water molecule traces. The produced FFAs were used as starting materials in decyl esters production by esterification reaction.

2.4. Preparation and functionalization of RHS particles

Octyl–SiO_2 and Phe–SiO_2 were produced via sequential SiO_2 extraction from a thermochemical process of rice husks and functionalization with OCTES and PheTES in toluene media at 120 °C according previous studies conducted in our lab [46,47].

2.5. Preparation of the heterogeneous biocatalysts using different RHS particles

TLL was immobilized via physical adsorption at low ionic strength to prevent lipase-lipase aggregation (5 mmol L^{-1}) buffer sodium acetate pH 5.0 at a ratio between support and enzyme solution (1:20), 25 °C, and 15 h of contact time under mechanical stirring (200 prm) by offering a fixed protein amount of 40 mg g^{-1} of support [46,47,69]. TLL adsorption process was monitored by measuring the disappeared protein amount in the supernatant solution according to Bradford's method [70], using bovine serum albumin as standard. The hydrolytic activity (HA) of the immobilized biocatalysts was determined on the hydrolysis of olive oil emulsion [46,47]. One international unit (1 U) was defined as the mass of biocatalyst required to release 1 μmol of FFA per minute of reaction at pH 8.0 and 37 °C. Specific activity (SA) was calculated by the ratio between hydrolytic activity (HA) and protein concentration.

2.6. Decyl esters production via esterification in solvent-free systems

Decyl esters were produced by esterification reaction of the FFA obtained by enzymatic hydrolysis of soybean-base oils with decanol in solvent-free systems. The reactions were performed in closed flasks (100 mL) containing 18.3 mmol of each reactant (5.1 g of FFA + 2.9 g of decanol) immersed in a thermostatic water bath shaker under constant stirring (240 rpm) at 40 ± 1 °C. Next, the biocatalysts were added to the reaction mixtures at a proportion of 40 mg protein (or 5 mg protein g^{-1} of reaction mixture) that corresponds to 1.5 g of TLL–SiO_2, 1.8 g of TLL–Octyl–SiO_2, 1.45 g of TLL–Phe–SiO_2, or 0.7 g of Lipozyme TL IM. The reaction was also conducted using liquid lipase extract under the same conditions and protein concentration (2.32 mL of liquid lipase extract). Samples from the reaction mixture (50 μL) were periodically withdrawn and titrated with NaOH solution (30 mmol L^{-1}) to determine FFA conversion – Y (%), as shown in Eq. (2) [22,28]. The reactions were performed with two replications and the experimental data are represented as mean ± s.d.

$$Y = \left(\frac{FFA_0 - FFA_t}{FFA_0}\right) \times 100 \quad (2)$$

where FFA_0 is the initial FFA concentration (mol L^{-1}), and FFA_t is the residual FFA concentration after a certain time t (mol L^{-1}).

The productivity (P – μmol min^{-1} $mg_{protein}^{-1}$), based on the FFA consumption, was estimated as follows (Eq. (3)):

$$P = \frac{(FFA_0 - FFA_e) \times V}{t_e \times m} \quad (3)$$

where: V is the volume of reaction mixture (≈9.2 × 10^{-3}L), FFA_0 is the initial FFA concentration (mol L^{-1}), FFA_e is the residual FFA concentration at equilibrium (mol L^{-1}), t_e is the reaction time necessary to achieve the chemical equilibrium (min), and m is the initial protein amount in the reaction (40 mg).

2.6.1. Biocatalysts operational stability

The operational stabilities of the biocatalysts were investigated by performing fifteen consecutive esterification batches using both FFA from RSBO and USCO under the conditions described in Section 2.6. After each esterification reaction, the heterogeneous biocatalysts (TLL–SiO_2 or TLL–Octyl–SiO_2) were recovered by vacuum filtration, followed by washing with cold hexane and drying at 25 °C by 2 h. Subsequently, they were reintroduced in fresh reaction mixtures to start new reaction batches.

2.7. Esters purification

The heterogeneous biocatalysts were separated from the reaction mixture via vacuum filtration. The produced decyl esters were purified in silica gel packed in Waters Sep-Pak® Vac cartridges (10 g) using hexane:ethyl acetate gradient elution (9:1, 8:2, 7:3 and 1:1 v v^{-1}) as mobile phase. The fractions with decyl esters were collected and the mobile phase solvent was evaporated using a rotating bath under vacuum at 50 °C. The esters were dried overnight at 25 °C using anhydrous sodium sulfate at a concentration of 20% m v^{-1}.

2.8. NMR analysis

Proton nuclear magnetic resonance (^{1}H NMR) spectra were recorded on a Bruker AC 300 spectrometer (Bruker BioSpin, Germany) at a frequency of 300 MHz. Samples of the purified esters (10 mg) were dissolved in 500 μL of deuterated chloroform ($CDCl_3$) containing tetramethylsilane (TMS) as standard.

2.9. TGA/DSC analysis

TGA and DSC curves were obtained using an SDT-Q600 analyzer (TA Instruments) operating at a heating rate of 10 °C min^{-1} and an air flow at 100 mL min^{-1} using a mass of around 7.0 mg for each ester loaded in an alumina sample holder.

2.10. Physicochemical analyses

The properties of the both oleaginous feedstocks (RSBO and USCO) and decyl esters were such as kinematic viscosity at 40 and 100 °C (ASTM D445), viscosity index (ASTM D2270), acid value (ASTM D947), moisture content (ASTM D7546–15), saponification value (ASTM D464–15) and density at 23.5 °C (ASTM D891) were determined according to standard methods – American Society for Testing and Materials [71].

3. Results and discussion

3.1. Feedstocks physicochemical characterization

The physicochemical properties of the oleaginous feedstocks were firstly determined. Saponification values for both RSBO and USCO were very close, 189.1 and 192.3 mg_{KOH} g^{-1} of oil, respectively. Exceptionally for USCO, similar result was reported in a previous study [9]. According to results, frying process led to an increase of kinematic viscosity of soybean oil from 31.1 to 32.2 mm^2 s^{-1} due to the oxidative degradation of double bonds from the polyunsaturated fatty acids, mainly linoleic and linolenic acids [72], thus resulting in undesirable reactions such as hydrolysis, interesterification, and polymerization. Moreover, moisture of USCO (0.8% m m^{-1}) can contributes for its degradation by hydrolysis to liberate FFA [7,72]. This could cause an increase of viscosity, as aforementioned, and acid value – AV [72]. According to these results, AV was raised from 0.2 to 0.9 mg_{KOH} g^{-1} of oil after frying process, which confirms a possible degradation of soybean oil by hydrolysis reaction, as expected. This indicates that the biolubricant production via conventional homogeneous transesterification using alkaline metal

hydroxides and/or alkoxides could be recommended only for refined feedstock (RSBO), since FFA and water molecules present in USCO could react with alkaline catalysts to produce soaps that cause serious purification difficulties [72–75].

3.2. Enzymatic hydrolysis of RSBO: Influence of relevant parameters on the FFA production

In this set of experiments, the influence of CRL loading, reaction temperature and mechanical stirring speed on the hydrolysis reactions was investigated. The selection of these parameters was due to their reported importance on the enzymatic hydrolysis of several vegetable oils [27,28,76]. In this study, oil hydrolysis reactions were performed in emulsifier-free systems and absence of buffering agents, which make an interesting strategy from the environmental and economic aspects, since fewer operational steps in the separation/purification of FFA are required.

3.2.1. Influence of CRL concentration

Initially, it was evaluated the influence of CRL loading varying from 30 U g^{-1} to 60 U g^{-1} of reaction mixture, that corresponds respectively to 1.85 mg_{powder} g^{-1} and 3.7 mg_{powder} g^{-1} of reaction mixture, in order to determine the minimum biocatalyst concentration required for an effective reaction. According to Fig. 2A, a raise in CRL concentration in the reaction mixture from 30 to 50 U g^{-1} led to an increase in hydrolysis percentage from 61.5% (at 150 min of reaction) to 81.3% (at 240 min of reaction). On the other hand, further increase in CRL concentration (60 U g^{-1}) do not produce a significant improvement on the hydrolysis percentage (83.5% at 240 min of reaction). These results suggest a possible saturation of oil droplets with CRL molecules above 50 U g^{-1} of reaction mixture. Therefore, an initial CRL loading of 50 U g^{-1} was enough to achieve maximum FFA production and, thus, chosen for further assays.

3.2.2. Influence of temperature

The temperature should produce an increment on the enzyme activity until reaching a point where the enzyme becomes inactivated [27,28]. In this study, the hydrolysis reactions were performed under different reaction temperature (from 25 °C to 50 °C) for different reaction times at a constant initial CRL loading (50 U g^{-1} of reaction mixture). As can be observed in Fig. 2B, increasing the temperature from 25 °C to 40 °C led to an improvement of the FFA. Under these conditions, hydrolysis percentage varied from 74.7% to 92.2% after 210 min and 240 min, respectively. However, further increase in reaction temperature resulted in a strong decrease of the catalytic activity of CRL due to its possible thermal inactivation, as aforementioned. As a result, a decrease of hydrolysis percentage above 40 °C was observed – maximum hydrolysis percentage of 85.2% and 72.3% after 210 min of reaction at 45 °C and 50 °C, respectively. Thus, 40 °C was selected for further tests.

3.2.3. Influence of stirring speed

Since the enzymatic hydrolysis of several TAG is strongly influenced through interface between oil and aqueous phases, an intense stirring speed may be required to reduce the oil droplets size to increase the oil drops interfacial area, thus improving the FFA production [28]. The influence of mechanical stirring speed from 500 to 2000 rpm was

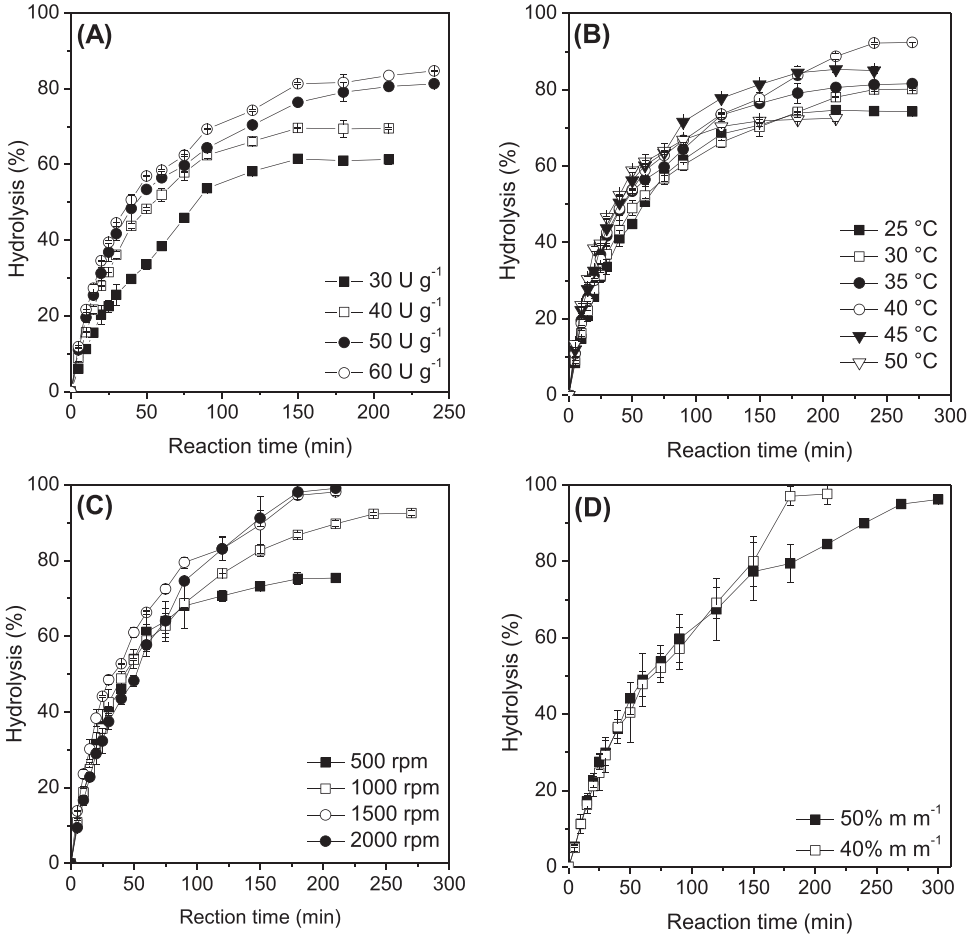


Fig. 2. Influence of CRL loading (A), temperature (B) and mechanical stirring (C) on the hydrolysis of RSBO using a ratio oil:water of 50% (m m^{-1}). Influence of oil: water mass ratio on FFA production from USCO performed at 40 °C, 1500 rpm and CRL loading of 50 U g^{-1} of reaction mixture (D).

studied under fixed experimental conditions – 40 °C and initial CRL loading of 50 U g^{-1} (Fig. 2C). Preliminary study under static condition was performed, however FFA production was not observed (data not shown). As expected, an increase in mechanical stirring frequency from 500 to 1500 rpm had a positive influence on the FFA production, very likely by increasing the oil/water interface surface, that way enabling the full activity of all lipase molecules. Under such experimental conditions, hydrolysis percentage varied from 75.3% (500 rpm and 210 min of reaction) to ≈ 98% (1500 rpm and 180 min of reaction) – see Fig. 2C. Similar result was observed at highest mechanical stirring speed (2000 rpm) – 98.5% after 180 min of reaction. Based on these results, 1500 rpm was chosen to produce FFA.

3.2.4. FFA production using waste soybean cooking oil as feedstock

The experimental conditions that promotes the maximum FFA production from RSBO was also used for waste oil (USCO) hydrolysis – 40 °C, initial CRL loading of 50 U g^{-1}, mechanical stirring speed of 1500 rpm and reaction mixture consisting of a ratio oil:water of 50% (m m^{-1}). According to Fig. 2D, a hydrolysis percentage of 85% after 180 min of reaction was obtained under such operating conditions. Therefore, almost full conversion (around of 99%) into FFA by using USCO as feedstock was observed using a ratio oil:water of 40% (m m^{-1}). This is due to its higher viscosity than RSBO (see Table 1) which could reduce the dispersion of oil droplets, reducing their interfacial area and, thus, their capacity to adsorb enzyme molecules. This reduces the catalytic activity of the enzyme [27,77]. Thus, a subsequent assay was conducted at an oil:water mass ratio of 40% (m m^{-1}) in order to obtain a reaction mixture with satisfactory dispersion of substrate and lipase molecules. This system enabled a better dispersion of oil droplets in the reaction mixture, thus confirming the strong influence of rheological properties of the reaction mixture. Under these conditions, almost full hydrolysis of USCO can be observed after 180 min of reaction, confirming the importance of oil droplets size on the hydrolysis of TAGs. Thus, FFA production via enzymatic hydrolysis of USCO was performed using oil:water mass ratio of 40% (m m^{-1}).

3.3. Characterization of the TLL-RHS biocatalysts

In this study, rice husk silica particles were chemically functionalized with OCTES and PheTES to introduce hydrophobic moieties (octyl and phenyl groups) on their surface. That way, TLL immobilization on both hydrophobic supports proceeds via interaction of hydrophobic pockets around its active centers in its open form (mechanism of interfacial activation). Thus, the TLL`s active centers are directed towards the support surface. The representative scheme of support functionalization and lipase immobilization is shown in Fig. 3. Their catalytic performances in decyl esters production (biolubricants) via esterification in solvent-free systems were compared with TLL immobilized on non-functionalized RHS (TLL–SiO_2). This more hydrophilic support has an isoelectric point (pI) between pH 1.0 and 2.0 [78,79]. Therefore, non-functionalized RHS presents a global anionic nature at pH 5.0 due to deprotonation of silanol groups (–Si–O^-) in its surface [80]. Under such condition (pH 5.0), TLL is also a weak anion, since its pI is pH 4.4 [62].

Table 1
Immobilization parameters of TLL immobilized via physical adsorption on differently activated rice husk silica.

Biocatalysts	Immobilization parameters		
	Protein (mg g^{-1})	HA[a] (U g^{-1})	SA[b] (U $mg_{protein}^{-1}$)
Soluble TLL	16.0	27,302.4	1706.8
TLL–SiO_2	26.8 ± 2.2	784.1 ± 49.9	29.2
TLL–Phe–SiO_2	27.7 ± 2.3	655.9 ± 31.9	23.7
TLL–Octyl–SiO_2	21.3 ± 0.4	618.9 ± 4.9	29.1

[a] : Hydrolytic activity
[b] : Specific activity

However, terminal amino group (pK between 6.5 and 7.0) and amino groups from surface lysine (pK above 10.7) or arginine residues of TLL are in their protonated forms (–NH_3^+). A high density of these groups can be observed in the opposite region to its active centers [62], perhaps enabling the ion exchange of the enzyme by this region with the support surface. It should be considered that ion exchange require the possibility of performing a multipoint ionic interaction between the support and the enzyme, not being a requirement to present opposite charges [81]. Thus, the adsorption process may occur via ionic interactions of negative groups of non-functionalized RHS (SiO_2) with positive groups of TLL (protonated amino groups). Therefore, lipase molecules adsorbed on this support can be oriented towards the reaction medium (see Fig. 3).

In fact, the different prepared supports provided heterogeneous biocatalysts with different immobilized protein concentrations and hydrolytic activity, as shown in Table 1. Both, non-functionalized RHS (SiO_2) and Phe–SiO_2 exhibited a similar maximal immobilized protein concentration (around 28 mg g^{-1} of support), while for Octyl–SiO_2, the maximum protein loading decreased to ≈22 mg g^{-1}. Although different immobilized protein concentrations have been obtained using both functionalized supports, they exhibited a similar hydrolytic activity between 618.9 and 655.9 U g^{-1}. However, TLL–SiO_2 exhibited the highest hydrolytic activity value (784.1 ± 49.9 U g^{-1}). Similar specific activity values (from 23.7 to 29.2 U $mg_{protein}^{-1}$) for all heterogeneous biocatalysts can be observed. However, these values were much lower than those observed using soluble TLL (1706.8 U $mg_{protein}^{-1}$). These difference could be caused by possible mass transfer limitations that can cause pH or substrate gradients inside pores of the heterogeneous biocatalysts [82].

3.4. Enzymatic decyl esters production via esterification in solvent-free systems

The catalytic performance of soluble TLL and prepared heterogeneous biocatalysts was also evaluated in decyl esters production – Fig. 4. The catalytic performances of these biocatalysts were compared with the commercial biocatalyst (Lipozyme TL IM). The experimental FFA conversion and productivity for each TLL formulation, expressed in terms of FFA consumption, are summarized in Table 2. As expected, different catalytic performances for each biocatalyst can also be observed in decyl esters production. Moreover, their catalytic properties were also influenced by the FFA source. A similar FFA conversion percentage (between 81.3% and 87.6%) for both reaction systems was achieved using all prepared heterogeneous biocatalysts. According to Table 2, the catalytic performance of the commercial biocatalyst was very similar to TLL–Octyl–SiO_2 in terms of FFA conversion and productivity, thus showing its promising use as biocatalyst in decyl esters production due to its renewable and eco-friendly preparation using an agricultural waste as silica source (rice husks). It is also possible to observe in Table 2 that all heterogeneous biocatalysts used in this study (non- and commercial immobilized TLL) exhibited FFA conversion values almost 2.5-times higher than that of its free form – no immobilized TLL (maximum FFA conversion only of 32%). This could be because TLL in anhydrous media should be in aggregated form, presenting much higher diffusional problems than the enzyme molecules dispersed in the surface of the biocatalysts. These results show that a correct immobilization protocol can significantly improve the activity and stability of lipases [83,84]. Moreover, they show that the results obtained in a hydrolytic process cannot fit to those obtained in a synthetic process.

Moreover, the differently prepared heterogeneous biocatalysts presented very different performance, and the results did not fit those found in olive oil emulsion hydrolysis. As shown in Fig. 4 and Table 2, the assayed biocatalysts exhibited a higher catalytic activity for FFA obtained from USCO, with exception of TLL–Octyl–SiO_2 that presented similar catalytic activity for FFA from both sources (see productivity values – ratio between consumed FFA concentration and reaction time at equilibrium). TLL–Phe–SiO_2 was the least efficient biocatalyst for these

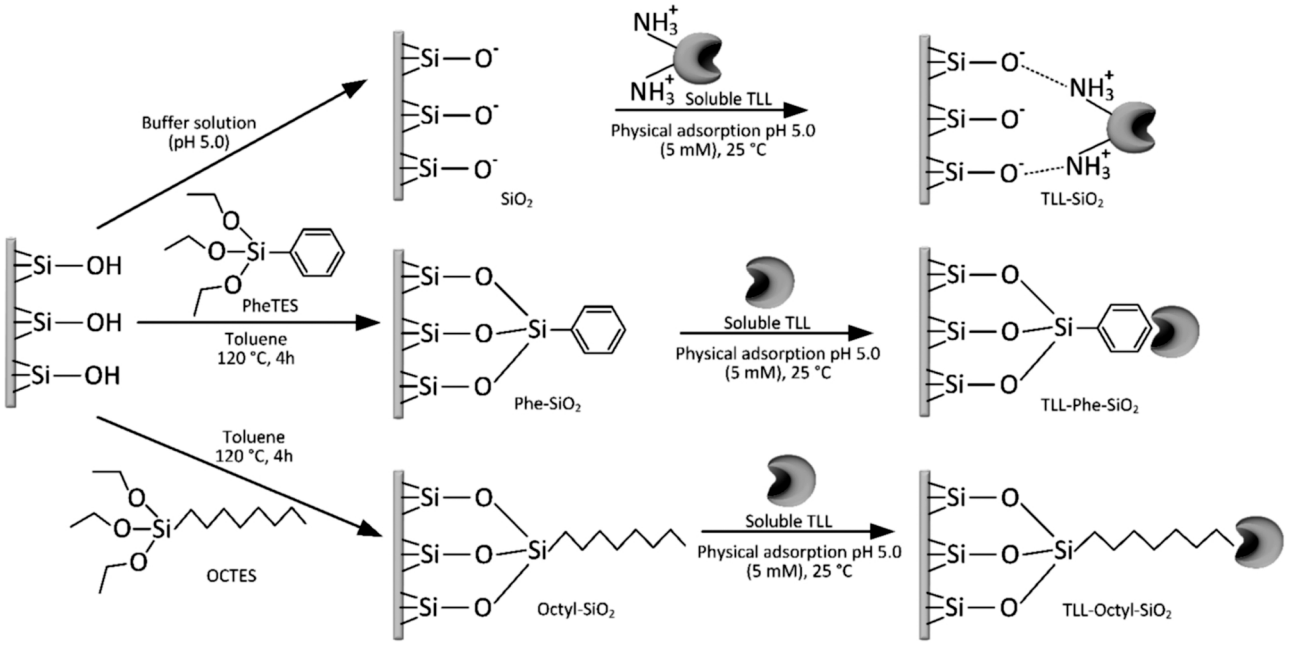


Fig. 3. Representative scheme of preparation and functionalization of RHS and TLL immobilization via physical adsorption.

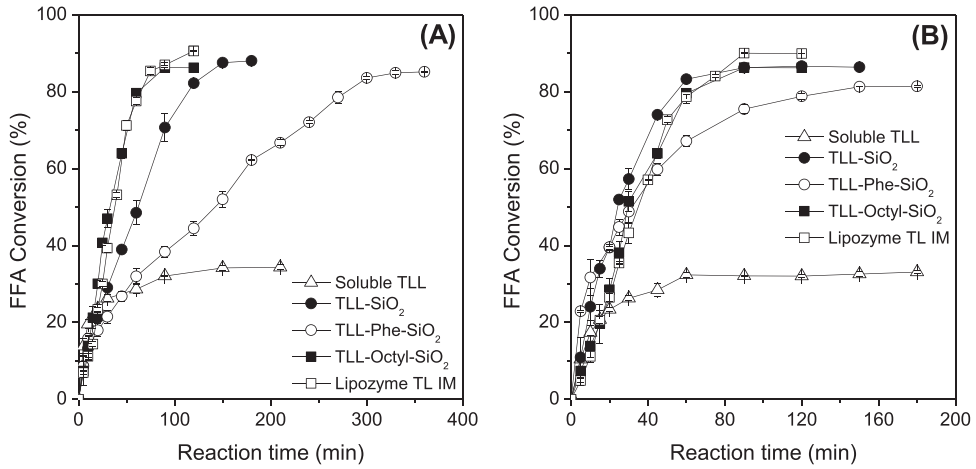


Fig. 4. Influence of reaction time in the enzymatic production of decyl esters using FFA from RSBO (A) and USCO (B). The esterification reactions were conducted in solvent-free systems at 40 °C, 240 rpm, stoichiometric FFA:decanol molar ratio (1:1), and fixed protein concentration (5 mg g^{-1} of reaction mixture).

Table 2
Influence of biocatalysts on the decyl esters production via direct esterification reaction.

Biocatalysts	RSBO			USCO		
	Conversion (%)	t_e[a] (min)	P[b] (μmol min^{-1} $mg_{protein}^{-1}$)	Conversion (%)	t_e[a] (min)	P[b] (μmol min^{-1} $mg_{protein}^{-1}$)
Soluble TLL	32.5	90	0.19	32.2	60	0.28
TLL–SiO_2	87.6	150	0.31	86.3	90	0.50
TLL–Phe–SiO_2	83.6	300	0.15	81.3	150	0.28
TLL–Octyl–SiO_2	86.0	90	0.51	86.5	90	0.53
Lipozyme TL IM	90.6	120	0.42	90.0	90	0.55

[a] : Reaction time at equilibrium
[b] : Productivity

reactions, although in olive oil emulsion hydrolysis it was among the most active ones. This heterogeneous biocatalyst gave the lowest productivity values among all prepared heterogeneous biocatalysts, which were very similar to those from the soluble lipase (see Table 2). The influence of RHS acyl layer on the features of immobilized lipases had been previously reported, and could be associated to different forms of TLL on the supports surface [85–87]. The highest productivities (≈0.5 μmol min^{-1} $mg_{protein}^{-1}$) were obtained for TLL–SiO_2 and Lipozyme TL IM using FFA from USCO and TLL–Octyl–SiO_2 for both FFA sources, followed by Lipozyme TL IM and TLL–SiO_2 using FFA from RSBO (0.42 and 0.31 μmol min^{-1} $mg_{protein}^{-1}$, respectively). Thus, TLL–Octyl–SiO_2, the most active prepared biocatalyst, and TLL–SiO_2 due

to its high catalytic activity in this reaction and ease preparation, since it does not require chemical functionalization steps, were chosen to perform operational stability (reusability) tests. TLL–Phe–SiO_2 was discarded due to its low activity in this process.

The catalytic performance of the selected heterogeneous biocatalyst (TLL–Octyl–SiO_2) prepared in this study was compared with previous studies described in the literature for the production of monoalkyl esters with lubricant properties. A two-step process (hydroesterification) via enzymatic hydrolysis of waste cooking oil (WCO) using free CRL and esterification of FFA with octanol catalyzed by a commercial acidic heterogeneous catalyst (Amberlyst 15 H) was proposed [88]. In this study, a FFA conversion of 100% was observed after 2.5 h of reaction using an excess of octanol (FFA:octanol molar ratio of 1:3), 80 °C (twice higher than that the present study – 40 °C), and presence of silica gel as a desiccant agent (50% of mass of FFA). In another study, octyl esters were also produced via esterification of FFA from WCO using Novozym 435 as biocatalyst [89], a commercial biocatalyst prepared via physical adsorption of Lipase B from *Candida antarctica* on Lewatit VP OC 1600 [90]. Under optimal conditions, FFA conversion of 100% after 3 h of reaction at 60 °C and FFA:octanol molar ratio of 1:2.5 was achieved. In a recent study, monoalkyl esters were also produced via enzymatic esterification of medium-chain fatty acids (caprylic, capric and lauric acids) with octanol catalyzed by five different heterogeneous biocatalysts (non-commercial and commercial biocatalysts) [91]. Maximum acid conversion above 90% were achieved at 24 h of reaction performed using an slight excess of octanol (acid:alcohol molar ratio de 1:1.3). These results show the promising use of the heterogeneous biocatalyst prepared in this study due to its high catalytic activity under moderate conditions (maximum FFA conversion very close to 90% after 90–120 min of reaction at 40 °C) and no requirement of excess of reactants to obtain high FFA conversion. This may facilitate the esters separation from the reaction mixture and minimizes wastes generation in the process.

3.5. Reusability tests

The reusability of the selected heterogeneous biocatalysts prepared in this study (TLL–SiO_2 and TLL–Octyl–SiO_2) was analyzed during fifteen consecutive esterification batches using both FFA sources. According to Fig. 5, TLL–Octyl–SiO_2 was much more stable than TLL–SiO_2: TLL–Octyl–SiO_2 presented a retention of its initial activity around 90% after 15 esterification batches using both FFA sources (FFA conversion percentage between 77.5% and 79% for both reaction systems). However, a gradual decrease of the catalytic activity during operation can be verified for TLL–SiO_2 using FFA from RSBO (Fig. 5A). At 15th esterification batch, FFA conversion was of 39%, that corresponds to 55% of its original activity. On the other hand, a drastic decrease of 90% of activity can be observed for decyl esters production using FFA from USCO and maximum FFA conversion only of 9.2% at the 15th batch was achieved (Fig. 5B). This shows that the RHS functionalization with octyl groups permits a much stronger adsorption of the enzyme and in a more stable form [46,92]. It is also possible to observe a higher decrease of activity of TLL–SiO_2 using USCO, perhaps due to possible the presence of some contaminants (e.g., generated by oxidation, including oxidized or polymerized FFA, as explained above). This could have an inactivating effect of the enzyme or even lead to obstruction of some pores of the biocatalyst or improve enzyme desorption. These results clearly demonstrate that the introduction of hydrophobic groups on the RHS is a crucial step for preparing a heterogeneous TLL biocatalyst with high stability after successive esterification batches.

3.6. Identification and characterization of physicochemical properties of decyl esters

The identification of the produced decyl esters from both FFA sources (RSBO and USCO) was performed by ^{1}H NMR analysis, as shown in Fig. 6. According to results, a triplet at 4.00 ppm can be observed in both spectra. This signal is due to hydrogen atoms of the methylene group attached to the oxygen of the ester (–CH_2–$COOCH_2$–). The signal at 2.25 ppm is characteristic of hydrogen atoms of the methylene group attached to the carboxyl group of the ester (–CH_2–$COOCH_2$–). These signals confirm the enzymatic production of decyl esters.

The physicochemical properties of the decyl esters produced in this study were determined according to standard methods (ASTM) and compared with those of the raw materials (RSBO and USCO) and commercial synthetic monoesters (biolubricants) supplied by Zschimmer & Schwarz Group (Table 3). The produced decyl esters from both FFA sources exhibited similar kinematic viscosity values at 40 (≈9 $mm^2\ s^{-1}$) and 100 ºC (≈3 $mm^2\ s^{-1}$). These values are in accordance with commercial synthetic monoesters. A decrease in viscosity index values from 223–227 for both oleaginous feedstocks (raw materials) to 190–204 after hydroesterification process can be observed. This decrease of viscosity values is due to reduction of molecular mass that results in a lowest stracking of adjacent molecules and, thus, intermolecular forces [93,94]. This means that both produced monoesters have lower resistance to temperature changes than that of their raw materials (RSBO and USCO). However, as can be shown in Table 3, both produced decyl esters had also viscosity index values very similar to the commercial

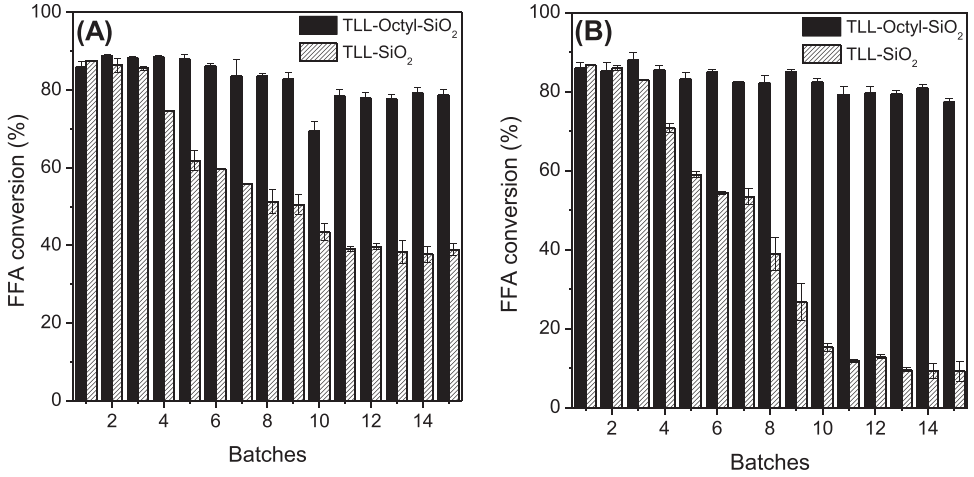


Fig. 5. Reusability tests after successive esterification of decanol with FFA from RSBO (A) and USCO (B) in solvent-free systems catalyzed by TLL–SiO_2 and TLL–Octyl–SiO_2. The esterification reactions were performed in solvent-free systems at 40 °C, 240 rpm, stoichiometric FFA:decanol molar ratio (1:1), and fixed protein concentration (5 mg g^{-1} of reaction mixture).

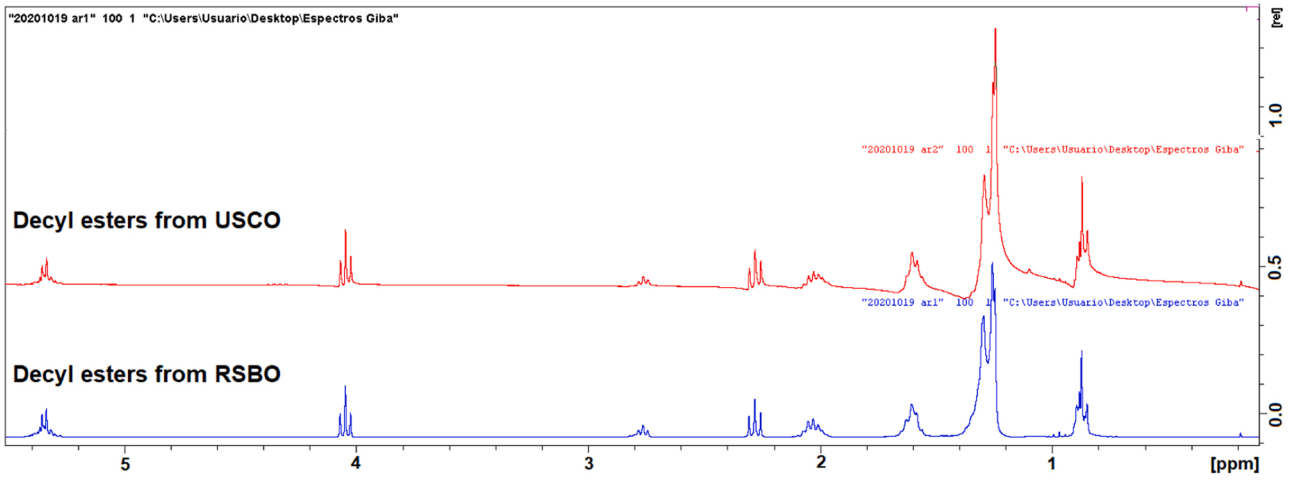


Fig. 6. ^{1}H NMR spectra of decyl esters from RSBO and USCO.

Table 3
Physicochemical properties of oleaginous feedstocks (RSBO and USCO) and produced decyl esters and their comparison with commercial biolubricant monoesters.

Properties	RSBO	USCO	Decyl esters from RSBO	Decyl esters from USCO	LEXOLUBE HS-C[a]	LEBRICIT 10 L[b]	LEXOLUBE T-110[c]
Kinematic viscosity at 40 °C ($mm^2\ s^{-1}$)	31.1	32.2	8.7	8.9	9.0	7.0	9.5
Kinematic viscosity at 100 °C ($mm^2\ s^{-1}$)	7.5	7.8	2.8	2.9	3.0	2.3	2.9
Viscosity index	223	227	190	204	220	170	170
Density at 23.5 °C (Kg m^{-3})	912	918	865	867	NR	NR	NR
Acid value ($mg_{KOH}\ g^{-1}$)	0.2	0.9	0.8	1.3	NR	NR	NR

[a] : Reference [101].
[b] : Reference [102].
[c] : Reference [103].

monoesters, thus showing their promising properties as lubricity additive, or friction modifier for a broad range of metalworking applications industrial biolubricants. Acid values for both produced esters varied from 0.8 $mg_{KOH}\ g^{-1}$ to 1.3 $mg_{KOH}\ g^{-1}$ using FFA from RSBO and USCO, respectively. It is also possible to observe a slight decrease of density values of the oils (912–918 Kg m^{-3}) after conversion into decyl esters (865–867 Kg m^{-3}). This decrease could be due to a possible increase of molecular distance between adjacent ester molecules by replacing glycerol moiety in the triacylglycerol structures by decyl groups. The decrease of density values in monoalkyl esters production from several triacylglycerols has also been observed in previous reports [95,96]. The density values of both decyl esters were very similar those reported for monoesters produced by transesterification of isopropanol or 2-ethyl-hexanol with ethyl esters from waste fish oil [19], transesterification of palm kernel oil with isoamyl alcohol or fusel oil [95], or monoesters from fusel oil and oleic acid [97].

3.7. Thermal analysis of the produced decyl esters

The thermal stability of both produced decyl esters from RSBO and USCO using TGA/DTG analysis was analyzed under oxidative atmosphere. As shown in Fig. 7, both prepared monoesters exhibited three stage mass losses between 18 °C and 1000 °C. According to Fig. 7A, the produced decyl esters from RSBO were thermally stable up to ≈ 200 °C. The produced decyl esters from USCO exhibited also a high thermal stability up to 200 °C. In this temperature range, a loss mass of only 3.26% can be observed due to possible presence of water traces in its composition (Fig. 7B). This high oxidative stability at this temperature range for both produced decyl esters was also confirmed by DTG analysis (see blue curves). It is possible to observe an intense loss mass of 78.7% and 82.7% for decyl esters from RSBO and USCO, respectively, from ≈ 200 °C to ≈ 350 °C due to degradation of ester bonds or double bounds existent in the chemical structure of mono- or polyunsaturated fatty acids (see fatty acid composition in Section 2.1). The mass loss above ≈ 350 °C (see also DTG curves for both decyl esters) suggests complete thermal degradation of the resulting oxidized fragments generated in the before stage (between ≈200 °C to ≈350 °C). These results show that both produced decyl esters exhibited high oxidative stability at a temperature range below 200 °C.

In this study, DSC analysis was also performed to determine the crystallization and melting points of the produced decyl esters. DSC curves of the esters from RSBO (A) and USCO (B) are shown in Fig. 8. Previous reports have demonstrated that the pour points of some biolubricants can be determined from the melting point values determined using DSC analysis [98,99]. This important parameter describes the lowest temperature at which a fluid can be poured under defined test conditions [1]. According to DSC thermograms for heating and cooling shown in Fig. 8, both produced decyl esters exhibited very similar crystallization and melting point values. These results show that the frying process had no strong influence on the crystallization and melting temperatures of the produced esters. It is possible to observe also that these esters present two endothermic peaks and two exothermic peaks due to their wide fatty acid composition: saturated fatty acids – palmitic and stearic (≈14% m m^{-1}); and mono- or polyunsaturated fatty acids – oleic, linoleic and linolenic (≈86% m m^{-1}) [27]. The two peaks during the cooling process (crystallization) at around 8 °C and −35 °C are due to presence of saturated and unsaturated fatty acids in both decyl esters composition. Two peaks at around −15 °C (unsaturated fatty acids) and 17 °C (saturated fatty acids) were also observed for the heating process – fusion. The influence of fatty acid composition on the crystallization and melting temperatures of the several synthetic esters with lubricant properties produced from vegetable or microbial oils has also been reported in other studies [22,100].

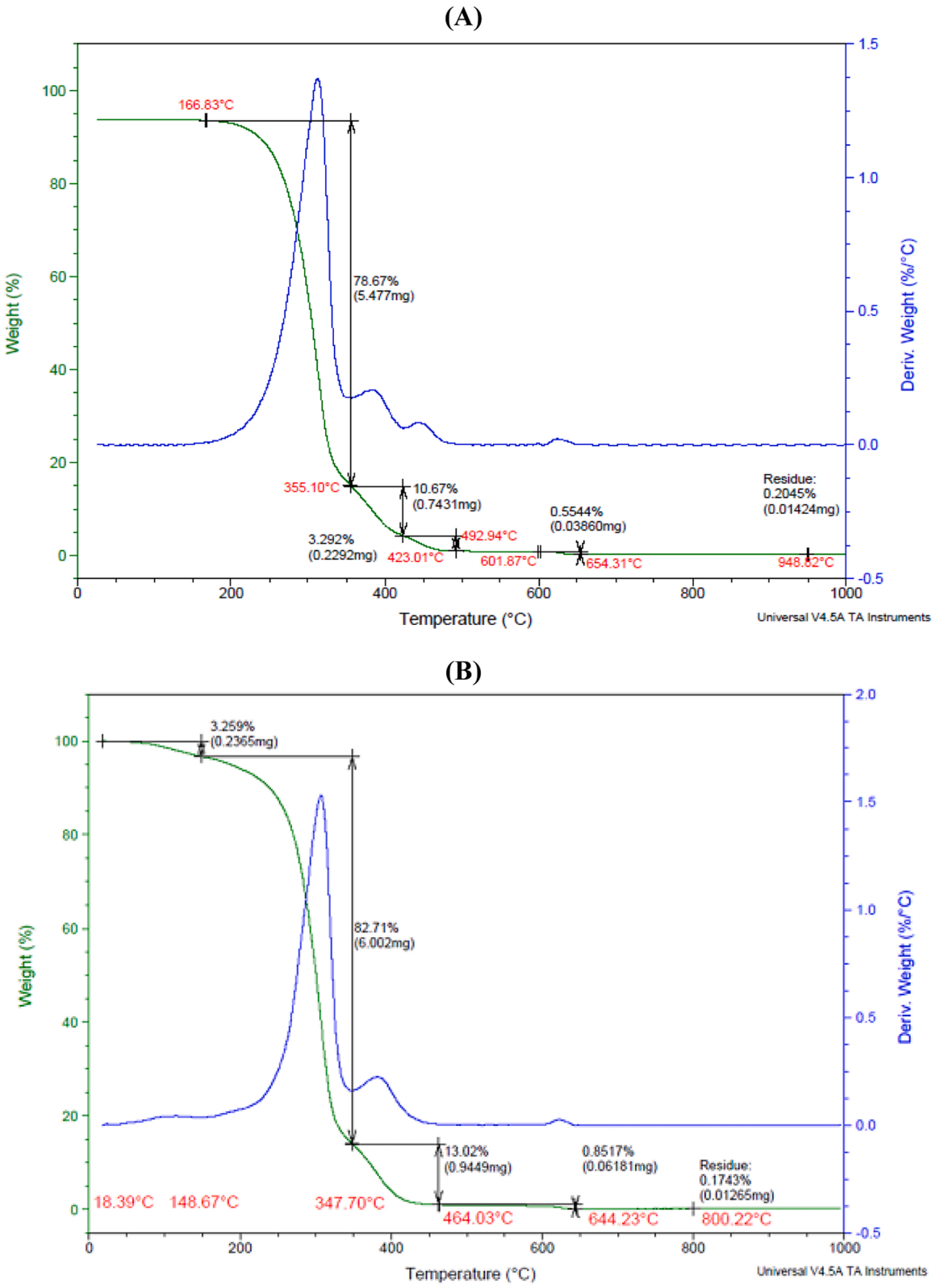


Fig. 7. TGA and DTG analyses of decyl esters from RSBO (A) and USCO (B).

4. Conclusion

This study reports a clean and sustainable process has been proposed to produce decyl esters in solvent-free systems via hydroesterification of both soybean oils. For such purpose, crude extract from CRL was used in FFA production. TLL (Lipolase® 100 L) immobilized on non- (TLL–SiO_2) and functionalized silica particles with phenyl (TLL–Phe–SiO_2) and octyl (TLL–Octyl–SiO_2) groups obtained from an agro-industrial waste (rice husk) were utilized as biocatalysts and their performance were compared with free lipase in decyl esters production via esterification reaction. The results obtained show that the TLL immobilization improved its performance catalytic in esters production using both FFA

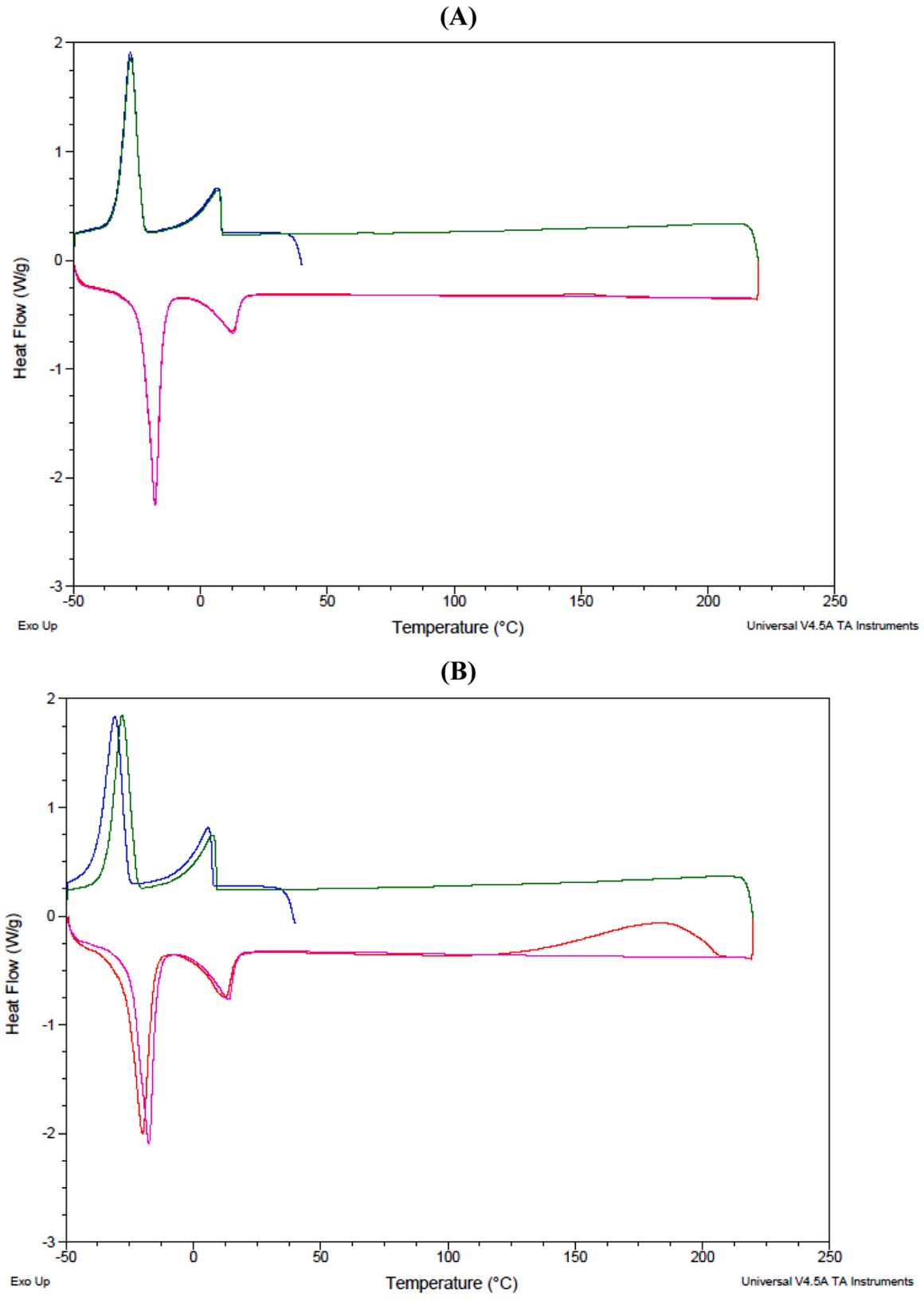


Fig. 8. DSC analyses of decyl esters from RSBO (A) and USCO (B).

sources. FFA conversion percentage between 81.3% and 87.6% was observed using immobilized lipase on both non- and functionalized supports, almost 2.5-times higher than free lipase. The highest productivity values were obtained only for TLL–SiO_2 and TLL–Octyl–SiO_2. Their catalytic performances were compared with a commercial biocatalyst (Lipozyme TL IM) and similar FFA conversion around ≈ 90% after 90–120 min of reaction was observed. Reusability tests using both FFA sources showed that the introduction of octyl groups in RHS surface was an important step to prepare biocatalysts with high operational stability. In fact, TLL–Octyl–SiO_2 exhibited a retention of its activity

above 90% after fifteen consecutive esterification reactions. The functionalization step prevented possible aqueous phase formation on the biocatalyst surface that can result in inactivation or desorption of the enzyme. The decyl esters produced using both FFA sources presented good oxidative stability below 200 °C and similar viscosity values comparable with commercial biolubricants, indicating their application as potential biolubricants.

CRediT authorship contribution statement

Guilherme J. Sabi: Visualization, Investigation, Data curation, Writing – original draft. **Rafaela S. Gama:** Visualization, Investigation, Data curation, Writing – original draft. **Roberto Fernandez-Lafuente:** Conceptualization, Funding acquisition, Writing – review & editing. **Juliana Cancino-Bernardi:** Conceptualization, Methodology, Writing – review & editing. **Adriano A. Mendes:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

Conflict of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

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References

[1] N.A. Zainal, N.W.M. Zulkifli, M. Gulzar, H.H. Masjuki, A review on the chemistry, production, and technological potential of bio-based lubricants, Renew. Sustain. Energy Rev. 82 (2018) 80–102, https://doi.org/10.1016/J.RSER.2017.09.004.

[2] Applications of Lubricants. Engineering360. ⟨https://www.globalspec.com/reference/38787/203279/Applications-of-Lubricants⟩ (Accessed 1 October 2021).

[3] Y. Singh, A. Sharma, A. Singla, Non-edible vegetable oil–based feedstocks capable of bio-lubricant production for automotive sector applications-a review, Environ. Sci. Pollut. Res. 26 (2019) 14867–14882, https://doi.org/10.1007/S11356-019-05000-9.

[4] G. Karmakar, P. Ghosh, B.K. Sharma, Chemically modifying vegetable oils to prepare green lubricants, Lubricant 5 (2017) 44, https://doi.org/10.3390/LUBRICANTS5040044.

[5] I.C.A. Bolina, R.A.B. Gomes, A.A. Mendes, Biolubricant production from several oleaginous feedstocks using lipases as catalysts: current scenario and future perspectives, BioEnergy Res. 14 (2021) 1039–1057, https://doi.org/10.1007/S12155-020-10242-4.

[6] Global vegetable oil consumption, 2019/20. Statista. ⟨https://www.statista.com/statistics/263937/vegetable-oils-global-consumption/⟩ (Accessed 15 September 2021).

[7] A.S. César, D.E. Werderits, G.L.O. Saraiva, R.C.S. Guabiroba, The potential of waste cooking oil as supply for the Brazilian biodiesel chain, Renew. Sustain. Energy Rev. 72 (2017) 246–253, https://doi.org/10.1016/j.rser.2016.11.240.

[8] S. Nanda, R. Rana, H.N. Hunter, Z. Fang, A.K. Dalai, J.A. Kozinski, Hydrothermal catalytic processing of waste cooking oil for hydrogen-rich syngas production, Chem. Eng. Sci. 195 (2019) 935–945, https://doi.org/10.1016/J.CES.2018.10.039.

[9] M. Sarno, M. Iuliano, C. Cirillo, Optimized procedure for the preparation of an enzymatic nanocatalyst to produce a bio-lubricant from waste cooking oil, Chem. Eng. J. 377 (2019), 120273, https://doi.org/10.1016/j.cej.2018.10.210.

[10] W.C.A. Carvalho, J.H.H. Luiz, R. Fernandez-Lafuente, D.B. Hirata, A.A. Mendes, Eco-friendly production of trimethylolpropane triesters from refined and used soybean cooking oils using an immobilized low-cost lipase (Eversa® Transform 2.0) as heterogeneous catalyst, Biomass Bioenergy 155 (2021), 106302, https://doi.org/10.1016/j.biombioe.2021.106302.

[11] S. Gryglewicz, M. Muszyński, J. Nowicki, Enzymatic synthesis of rapeseed oil-based lubricants, Ind. Crops Prod. 45 (2013) 25–29, https://doi.org/10.1016/j.indcrop.2012.11.038.

[12] M. Bahadi, J. Salimon, D. Derawi, Synthesis of di-trimethylolpropane tetraester-based biolubricant from *Elaeis guineensis* kernel oil via homogeneous acid-catalyzed transesterification, Renew. Energy 171 (2021) 981–993, https://doi.org/10.1016/J.RENENE.2021.02.163.

[13] M. Gul, N.W.M. Zulkifli, H.H. Masjuki, M.A. Kalam, M.A. Mujtaba, M.H. Harith, A.Z. Syahir, W. Ahmed, A. Bari Farooq, Effect of TMP-based-cottonseed oil-biolubricant blends on tribological behavior of cylinder liner-piston ring combinations, Fuel 278 (2020), 118242, https://doi.org/10.1016/J.FUEL.2020.118242.

[14] N.A.M. Aziz, H.A. Hamid, R. Yunus, Z. Abbas, R. Omar, U. Rashid, A.M. Syam, Kinetics and thermodynamics of synthesis of palm oil-based trimethylolpropane triester using microwave irradiation, J. Saudi Chem. Soc. 24 (2020) 552–566, https://doi.org/10.1016/J.JSCS.2020.05.006.

[15] P.A.B. Diaz, F. de, A. Kronemberger, A.C. Habert, A pervaporation-assisted bioreactor to enhance efficiency in the synthesis of a novel biolubricant based on the enzymatic transesterification of a castor oil based biodiesel, Fuel 204 (2017) 98–105, https://doi.org/10.1016/J.FUEL.2017.05.025.

[16] Z. Cai, X. Zhuang, X. Yang, F. Huang, Y. Wang, Y. Li, Litsea cubeba kernel oil as a promising new medium-chain saturated fatty acid feedstock for biolubricant base oil synthesis, Ind. Crops Prod. 167 (2021), 113564, https://doi.org/10.1016/J.INDCROP.2021.113564.

[17] J. Chen, X. Bian, G. Rapp, J. Lang, A. Montoya, R. Trethowan, B. Bouyssiere, J. F. Portha, J.N. Jaubert, P. Pratt, L. Coniglio, From ethyl biodiesel to biolubricants: options for an Indian mustard integrated biorefinery toward a green and circular economy, Ind. Crops Prod. 137 (2019) 597–614, https://doi.org/10.1016/J.INDCROP.2019.04.041.

[18] B. Angulo, J.M. Fraile, L. Gil, C.I. Herrerías, Bio-lubricants production from fish oil residue by transesterification with trimethylolpropane, J. Clean. Prod. 202 (2018) 81–87, https://doi.org/10.1016/J.JCLEPRO.2018.07.260.

[19] B. Angulo, M. Fraile, L. Gil, C.I. Herrerías, S.G. Sl, P.I.El Zafranar, Comparison of chemical and enzymatic methods for the transesterification of waste fish oil fatty ethyl esters with different alcohols, ACS Omega 5 (2020) 1479–1487, https://doi.org/10.1021/acsomega.9b03147.

[20] J. Greco-Duarte, E.D. Cavalcanti-Oliveira, J.A.C. Da Silva, R. Fernandez-Lafuente, D.M.G. Freire, Two-step enzymatic production of environmentally friendly biolubricants using castor oil: enzyme selection and product characterization, Fuel 202 (2017) 196–205, https://doi.org/10.1016/J.FUEL.2017.04.036.

[21] E.D.C. Cavalcanti, É.C.G. Aguieiras, P.R. da Silva, J.G. Duarte, E.P. Cipolatti, R. Fernandez-Lafuente, J.A.C. da Silva, D.M.G. Freire, Improved production of biolubricants from soybean oil and different polyols via esterification reaction catalyzed by immobilized lipase from *Candida rugosa*, Fuel 215 (2018) 705–713, https://doi.org/10.1016/j.fuel.2017.11.119.

[22] A. Papadaki, K.V. Fernandes, A. Chatzifragkou, E.C.G. Aguieiras, J.A.C. da Silva, R. Fernandez-Lafuente, S. Papanikolaou, A. Koutinas, D.M.G. Freire, Bioprocess development for biolubricant production using microbial oil derived via fermentation from confectionery industry wastes, Bioresour. Technol. 267 (2018) 311–318, https://doi.org/10.1016/j.biortech.2018.07.016.

[23] J.G.E. Guedes Júnior, F.R. Mattos, G.J. Sabi, W.C.A. Carvalho, J.H.H. Luiz, É. C. Cren, R. Fernandez-Lafuente, A.A. Mendes, Design of a sustainable process for enzymatic production of ethylene glycol diesters via hydroesterification of used soybean cooking oil, J. Environ. Chem. Eng. 10 (2022), 107062, https://doi.org/10.1016/J.JECE.2021.107062.

[24] M.S. Barbosa, C.C.C. Freire, L.M.S. Brandão, E.B. Pereira, A.A. Mendes, M. M. Pereira, Á.S. Lima, C.M.F. Soares, Biolubricant production under zero-waste *Moringa oleifera* Lam biorefinery approach for boosting circular economy, Ind. Crops Prod. 167 (2021), 113542, https://doi.org/10.1016/J.INDCROP.2021.113542.

[25] H. Pourzolfaghar, F. Abnisa, W.M.A.W. Daud, M.K. Aroua, A review of the enzymatic hydroesterification process for biodiesel production, Renew. Sustain. Energy Rev. 61 (2016) 245–257, https://doi.org/10.1016/j.rser.2016.03.048.

[26] M.H.M. Avelar, D.M.J. Cassimiro, K.C. Santos, R.C.C. Domingues, H.F. De Castro, A.A. Mendes, Hydrolysis of vegetable oils catalyzed by lipase extract powder from dormant castor bean seeds, Ind. Crops Prod. 44 (2013) 452–458, https://doi.org/10.1016/j.indcrop.2012.10.011.

[27] K.C. Santos, D.M.J. Cassimiro, M.H.M. Avelar, D.B. Hirata, H.F. de Castro, R. Fernández-Lafuente, A.A. Mendes, Characterization of the catalytic properties of lipases from plant seeds for the production of concentrated fatty acids from different vegetable oils, Ind. Crops Prod. 49 (2013) 462–470, https://doi.org/10.1016/j.indcrop.2013.05.035.

[28] A.P.P. Bressani, K.C.A. Garcia, D.B. Hirata, A.A. Mendes, Production of alkyl esters from macaw palm oil by a sequential hydrolysis/esterification process using heterogeneous biocatalysts: Optimization by response surface methodology, Bioprocess Biosyst. Eng. 38 (2015) 287–297, https://doi.org/10.1007/s00449-014-1267-5.

[29] S. Arana-Peña, D. Carballares, R. Morellon-Sterlling, Á. Berenguer-Murcia, A. R. Alcántara, R.C. Rodrigues, R. Fernandez-Lafuente, Enzyme co-immobilization: Always the biocatalyst designers' choice…or not? Biotechnol. Adv. 51 (2021), 107584 https://doi.org/10.1016/J.BIOTECHADV.2020.107584.

[30] C.O. Akerman, Y. Gaber, N.A. Ghani, M. Lämsä, R. Hatti-Kaul, Clean synthesis of biolubricants for low temperature applications using heterogeneous catalysts, J. Mol. Catal. B Enzym. 72 (2011) 263–269, https://doi.org/10.1016/j.molcatb.2011.06.014.

[31] O. Mowla, E. Kennedy, M. Stockenhuber, *In-situ* FTIR study on the mechanism of both steps of zeolite-catalysed hydroesterification reaction in the context of biodiesel manufacturing, Fuel 232 (2018) 12–26, https://doi.org/10.1016/J.FUEL.2018.05.096.

[32] M. Bilal, C.D. Fernandes, T. Mehmood, F. Nadeem, Q. Tabassam, L.F.R. Ferreira, Immobilized lipases-based nano-biocatalytic systems - a versatile platform with incredible biotechnological potential, Int. J. Biol. Macromol. 175 (2021) 108–122, https://doi.org/10.1016/J.IJBIOMAC.2021.02.010.

[33] N.S. Okura, G.J. Sabi, M.C. Crivellenti, R.A.B. Gomes, R. Fernandez-Lafuente, A. A. Mendes, Improved immobilization of lipase from *Thermomyces lanuginosus* on a new chitosan-based heterofunctional support: mixed ion exchange plus hydrophobic interactions, Int. J. Biol. Macromol. 163 (2020) 550–561, https://doi.org/10.1016/J.IJBIOMAC.2020.07.021.

[34] J. Gao, W. Kong, L. Zhou, Y. He, L. Ma, Y. Wang, L. Yin, Y. Jiang, Monodisperse core-shell magnetic organosilica nanoflowers with radial wrinkle for lipase immobilization, Chem. Eng. J. 309 (2017) 70–79, https://doi.org/10.1016/J.CEJ.2016.10.021.

[35] P. Adlercreutz, Immobilisation and application of lipases in organic media, Chem. Soc. Rev. 42 (2013) 6406–6436, https://doi.org/10.1039/c3cs35446f.

[36] A.R. Ismail, K.H. Baek, Lipase immobilization with support materials, preparation techniques, and applications: present and future aspects, Int. J. Biol. Macromol. 163 (2020) 1624–1639, https://doi.org/10.1016/j.ijbiomac.2020.09.021.

[37] L. Zhong, Y. Feng, G. Wang, Z. Wang, M. Bilal, H. Lv, S. Jia, J. Cui, Production and use of immobilized lipases in/on nanomaterials: a review from the waste to biodiesel production, Int. J. Biol. Macromol. 152 (2020) 207–222, https://doi.org/10.1016/j.ijbiomac.2020.02.258.

[38] E.A. Manoel, J.C.S. Santos, D.M.G. Freire, N. Rueda, R. Fernandez-Lafuente, Immobilization of lipases on hydrophobic supports involves the open form of the enzyme, Enzym. Microb. Technol. 71 (2015) 53–57, https://doi.org/10.1016/j.enzmictec.2015.02.001.

[39] R.C. Rodrigues, J.J. Virgen-Ortíz, J.C.S. Santos, Á. Berenguer-Murcia, A. R. Alcantara, O. Barbosa, C. Ortiz, R. Fernandez-Lafuente, Immobilization of lipases on hydrophobic supports: immobilization mechanism, advantages, problems, and solutions, Biotechnol. Adv. 37 (2019) 746–770, https://doi.org/10.1016/j.biotechadv.2019.04.003.

[40] R. Verger, "Interfacial activation" of lipases: facts and artifacts, Trends Biotechnol. 15 (1997) 32–38, https://doi.org/10.1016/S0167-7799(96)10064-0.

[41] R.D. Schmid, R. Verger, Lipases: interfacial enzymes with attractive applications, Angew. Chem. Int. Ed. 37 (1998) 1608–1633, https://doi.org/10.1002/(SICI)1521-3773(19980703)37:12<1608::AID-ANIE1608>3.0.CO;2-V.

[42] L. Sarda, P. Desnuelle, Action de la lipase pancréatique sur les esters en émulsion, Biochim. Biophys. Acta 30 (1958) 513–521, https://doi.org/10.1016/0006-3002(58)90097-0.

[43] M.D. Alves, F.M. Aracri, É.C. Cren, A.A. Mendes, Isotherm, kinetic, mechanism and thermodynamic studies of adsorption of a microbial lipase on a mesoporous and hydrophobic resin, Chem. Eng. J. 311 (2017) 1–12, https://doi.org/10.1016/J.CEJ.2016.11.069.

[44] F.A.P. Lage, J.J. Bassi, M.C.C. Corradini, L.M. Todero, J.H.H. Luiz, A.A. Mendes, Preparation of a biocatalyst via physical adsorption of lipase from *Thermomyces lanuginosus* on hydrophobic support to catalyze biolubricant synthesis by esterification reaction in a solvent-free system, Enzym. Microb. Technol. 84 (2016) 56–67, https://doi.org/10.1016/j.enzmictec.2015.12.007.

[45] P. Ngin, K. Cho, O. Han, Immobilization of soybean lipoxygenase on nanoporous rice husk silica by adsorption: retention of enzyme function and catalytic potential, Molecules 26 (2021) 291, https://doi.org/10.3390/MOLECULES26020291.

[46] N.B. Machado, J.P. Miguez, I.C.A. Bolina, A.B. Salviano, R.A.B. Gomes, O. L. Tavano, J.H.H. Luiz, P.W. Tardioli, É.C. Cren, A.A. Mendes, Preparation, functionalization and characterization of rice husk silica for lipase immobilization via adsorption, Enzym. Microb. Technol. 128 (2019) 9–21, https://doi.org/10.1016/J.ENZMICTEC.2019.05.001.

[47] R.S. Gama, I.C.A. Bolina, É.C. Cren, A.A. Mendes, A novel functionalized SiO_2-based support prepared from biomass waste for lipase adsorption, Mater. Chem. Phys. 234 (2019) 146–150, https://doi.org/10.1016/j.matchemphys.2019.06.002.

[48] V. Kasche, Mechanism and yields in enzyme catalysed equilibrium and kinetically controlled synthesis of β-lactam antibiotics, peptides and other condensation products, Enzym. Microb. Technol. 8 (1986) 4–16, https://doi.org/10.1016/0141-0229(86)90003-7.

[49] S.R. Marsden, L. Mestrom, D.G.G. McMillan, U. Hanefeld, Thermodynamically and kinetically controlled reactions in biocatalysis – from concepts to perspectives, ChemCatChem 12 (2020) 426–437, https://doi.org/10.1002/CCTC.201901589.

[50] E. Castillo, V. Dossat, A. Marty, J.S. Condoret, D. Combes, The role of silica gel in lipase-catalyzed esterification reactions of high-polar substrates, J. Am. Oil Chem. Soc. 74 (1997) 77–85, https://doi.org/10.1007/S11746-997-0148-3.

[51] V. Dossat, D. Combes, A. Marty, Continuous enzymatic transesterification of high oleic sunflower oil in a packed bed reactor: influence of the glycerol production, Enzym. Microb. Technol. 25 (1999) 194–200, https://doi.org/10.1016/S0141-0229(99)00026-5.

[52] A. Marty, V. Dossat, J.-S. Condoret, Continuous operation of lipase-catalyzed reactions in nonaqueous solvents: influence of the production of hydrophilic compounds, Biotechnol. Bioeng. 56 (1997) 232–237, https://doi.org/10.1002/(SICI)1097-0290(19971020)56:2<232::AID-BIT12>3.0.CO;2-I.

[53] E. Séverac, O. Galy, F. Turon, C.A. Pantel, J.S. Condoret, P. Monsan, A. Marty, Selection of CalB immobilization method to be used in continuous oil transesterification: analysis of the economical impact, Enzym. Microb. Technol. 48 (2011) 61–70, https://doi.org/10.1016/J.ENZMICTEC.2010.09.008.

[54] J.J. Virgen-Ortíz, V.G. Tacias-Pascacio, D.B. Hirata, B. Torrestiana-Sanchez, A. Rosales-Quintero, R. Fernandez-Lafuente, Relevance of substrates and products on the desorption of lipases physically adsorbed on hydrophobic supports, Enzym. Microb. Technol. 96 (2017) 30–35, https://doi.org/10.1016/J.ENZMICTEC.2016.09.010.

[55] D.B. Hirata, T.L. Albuquerque, N. Rueda, J.J. Virgen-Ortíz, V.G. Tacias-Pascacio, R. Fernandez-Lafuente, Evaluation of different immobilized lipases in transesterification reactions using tributyrin: advantages of the heterofunctional octyl agarose beads, J. Mol. Catal. B Enzym. 133 (2016) 117–123, https://doi.org/10.1016/J.MOLCATB.2016.08.008.

[56] A.B. Martins, J.L.R. Friedrich, J.C. Cavalheiro, C. Garcia-Galan, O. Barbosa, M.A. Z. Ayub, R. Fernandez-Lafuente, R.C. Rodrigues, Improved production of butyl butyrate with lipase from *Thermomyces lanuginosus* immobilized on styrene–divinylbenzene beads, Bioresour. Technol. 134 (2013) 417–422, https://doi.org/10.1016/J.BIORTECH.2013.02.052.

[57] N.G. Graebin, A.B. Martins, A.S.G. Lorenzoni, C. Garcia-Galan, R. Fernandez-Lafuente, M.A.Z. Ayub, R.C. Rodrigues, Immobilization of lipase B from *Candida antarctica* on porous styrene–divinylbenzene beads improves butyl acetate synthesis, Biotechnol. Prog. 28 (2012) 406–412, https://doi.org/10.1002/BTPR.1508.

[58] A.B. Martins, M.F. Schein, J.L.R. Friedrich, R. Fernandez-Lafuente, M.A.Z. Ayub, R.C. Rodrigues, Ultrasound-assisted butyl acetate synthesis catalyzed by Novozym 435: enhanced activity and operational stability, Ultrason. Sonochem. 20 (2013) 1155–1160, https://doi.org/10.1016/J.ULTSONCH.2013.01.018.

[59] N. Paludo, J.S. Alves, C. Altmann, M.A.Z. Ayub, R. Fernandez-Lafuente, R. C. Rodrigues, The combined use of ultrasound and molecular sieves improves the synthesis of ethyl butyrate catalyzed by immobilized *Thermomyces lanuginosus* lipase, Ultrason. Sonochem. 22 (2015) 89–94, https://doi.org/10.1016/J.ULTSONCH.2014.05.004.

[60] M.L. Verma, C.J. Barrow, M. Puri, Nanobiotechnology as a novel paradigm for enzyme immobilisation and stabilisation with potential applications in biodiesel production, Appl. Microbiol. Biotechnol. 97 (2013) 23–29, https://doi.org/10.1007/s00253-012-4535-9.

[61] R.R. Sousa, A.S. Silva, R. Fernandez-Lafuente, V.S. Ferreira-Leitão, Solvent-free esterifications mediated by immobilized lipases: a review from thermodynamic and kinetic perspectives, Catal. Sci. Technol. 11 (2021) 5696–5711, https://doi.org/10.1039/D1CY00696G.

[62] R. Fernandez-Lafuente, Lipase from *Thermomyces lanuginosus*: uses and prospects as an industrial biocatalyst, J. Mol. Catal. B Enzym. 62 (2010) 197–212, https://doi.org/10.1016/j.molcatb.2009.11.010.

[63] Lipozyme® TL IM for interesterified fats. Novozymes. ⟨https://biosolutions.novozymes.com/en/oils-fats/products/interestified-fats/lipozyme-tl-im⟩ (accessed February 13, 2022).

[64] P.A. Mendoza-Ortiz, R.S. Gama, O.C. Gómez, J.H.H. Luiz, R. Fernandez-Lafuente, E.C. Cren, A.A. Mendes, Sustainable enzymatic synthesis of a solketal ester - process optimization and evaluation of its antimicrobial activity, Catalysts 10 (2020) 218, https://doi.org/10.3390/catal10020218.

[65] J.W. Kim, B.H. Kim, Y. Kim, M.W. Lee, D.J. Im, I.H. Kim, Lipase-mediated synthesis of neopentyl glycol diester using a combination of reduced and standard pressure, J. Am. Oil Chem. Soc. 98 (2021) 1001–1007, https://doi.org/10.1002/AOCS.12532.

[66] S. Choi, B.H. Kim, D.S. No, S.W. Yoon, M.W. Lee, D.J. Im, I.H. Kim, Lipase-catalyzed synthesis of 2-ethylhexyl palmitate in a solvent free system using step changes in temperature, Biochem. Eng. J. 177 (2022), 108261, https://doi.org/10.1016/J.BEJ.2021.108261.

[67] A.N. Phan, T.M. Phan, Biodiesel production from waste cooking oils, Fuel 87 (2008) 3490–3496, https://doi.org/10.1016/j.fuel.2008.07.008.

[68] D. Rooney, L.R. Weatherley, The effect of reaction conditions upon lipase catalysed hydrolysis of high oleate sunflower oil in a stirred liquid-liquid reactor, Process Biochem 36 (2001) 947–953, https://doi.org/10.1016/S0032-9592(01)00130-3.

[69] J.J. Bassi, L.M. Todero, F.A.P. Lage, G.I. Khedy, J.D. Ducas, A.P. Custódio, M. A. Pinto, A.A. Mendes, Interfacial activation of lipases on hydrophobic support and application in the synthesis of a lubricant ester, Int. J. Biol. Macromol. 92 (2016) 900–909, https://doi.org/10.1016/j.ijbiomac.2016.07.097.

[70] M.M. Bradford, A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding, Anal. Biochem. 72 (1976) 248–254, https://doi.org/10.1016/0003-2697(76)90527-3.

[71] Section: 5 Petroleum Products, Lubricants, and Fossil Fuels - Ordering Options. ⟨https://www.astm.org/BOOKSTORE/BOS/section5.htm⟩ (accessed October 19, 2021).

[72] A.K. Paul, V.B. Borugadda, M.S. Bhalerao, V.V. Goud, *In situ* epoxidation of waste soybean cooking oil for synthesis of biolubricant basestock: a process parameter optimization and comparison with RSM, ANN, and GA, Can. J. Chem. Eng. 96 (2018) 1451–1461, https://doi.org/10.1002/CJCE.23091.

[73] J. Cvengroš, Z. Cvengrošová, Used frying oils and fats and their utilization in the production of methyl esters of higher fatty acids, Biomass-.-. Bioenergy 27 (2004) 173–181, https://doi.org/10.1016/J.BIOMBIOE.2003.11.006.
[74] C.C.A. Loures, M.S. Amaral, P.C.M. Da Rós, S.M.F.E. Zorn, H.F. Castro, M.B. Silva, Simultaneous esterification and transesterification of microbial oil from *Chlorella minutissima* by acid catalysis route: a comparison between homogeneous and heterogeneous catalysts, Fuel 211 (2018) 261–268, https://doi.org/10.1016/J.FUEL.2017.09.073.
[75] L.K. Santos, R.R. Hatanaka, J.E. Oliveira, D.L. Flumignan, Experimental factorial design on hydroesterification of waste cooking oil by subcritical conditions for biodiesel production, Renew. Energy 114 (2017) 574–580, https://doi.org/10.1016/J.RENENE.2017.07.066.
[76] M.S. Barbosa, C.C.C. Freire, L.C. Almeida, L.S. Freitas, R.L. Souza, E.B. Pereira, A. A. Mendes, M.M. Pereira, Á.S. Lima, C.M.F. Soares, Optimization of the enzymatic hydrolysis of *Moringa oleifera* Lam oil using molecular docking analysis for fatty acid specificity, Biotechnol. Appl. Biochem. 66 (2019) 823–832, https://doi.org/10.1002/BAB.1793.
[77] Bailey's Industrial Oil and Fat Products, Bailey's Ind. Oil Fat Prod. 2005. ⟨https://doi.org/10.1002/047167849X⟩.
[78] T.G. Pineda-Vásquez, A.E. Casas-Botero, M.E. Ramírez-Carmona, M.M. Torres-Taborda, C.H.L. Soares, D. Hotza, Biogeneration of silica nanoparticles from rice husk ash using *Fusarium oxysporum* in two different growth media, Ind. Eng. Chem. Res. 53 (2014) 6959–6965, https://doi.org/10.1021/IE404318W.
[79] W.K. Setiawan, K.Y. Chiang, Eco-friendly rice husk pre-treatment for preparing biogenic silica: gluconic acid and citric acid comparative study, Chemosphere 279 (2021), 130541, https://doi.org/10.1016/J.CHEMOSPHERE.2021.130541.
[80] W. Wolak, A.B. Kolomeisky, M.R. Dudek, M. Marć, L. Najder-Kozdrowska, Enhancing silica surface deprotonation by using magnetic nanoparticles as heating agents, J. Phys. D. Appl. Phys. 52 (2019), 465001, https://doi.org/10.1088/1361-6463/AB39FF.
[81] M. Fuentes, P. Batalla, V. Grazu, B.C.C. Pessela, C. Mateo, T. Montes, J. A. Hermoso, J.M. Guisan, R. Fernandez-Lafuente, Mixed ion exchange supports as useful ion exchangers for protein purification: purification of Penicillin G Acylase from *Escherichia coli*, Biomacromolecules 8 (2007) 703–707, https://doi.org/10.1021/BM060992M.
[82] W. Tischer, V. Kasche, Immobilized enzymes: crystals or carriers? Trends Biotechnol. 17 (1999) 326–335, https://doi.org/10.1016/S0167-7799(99)01322-0.
[83] S. Arana-Peña, N.S. Rios, D. Carballares, L.R.B. Gonçalves, R. Fernandez-Lafuente, Immobilization of lipases via interfacial activation on hydrophobic supports: Production of biocatalysts libraries by altering the immobilization conditions, Catal. Today 362 (2021) 130–140, https://doi.org/10.1016/j.cattod.2020.03.059.
[84] J.A. Martínez-Sanchez, S. Arana-Peña, D. Carballares, M. Yates, C. Otero, R. Fernandez-Lafuente, Immobilized biocatalysts of Eversa® Transform 2.0 and lipase from *Thermomyces lanuginosus*: comparison of some properties and performance in biodiesel production, Catalysts 10 (2020) 738, https://doi.org/10.3390/CATAL10070738.
[85] G. Fernandez-Lorente, C.A. Godoy, A.A. Mendes, F. Lopez-Gallego, V. Grazu, B. de las Rivas, J.M. Palomo, J. Hermoso, R. Fernandez-Lafuente, J.M. Guisan, Solid-phase chemical amination of a lipase from *Bacillus thermocatenulatus* to improve its stabilization via covalent immobilization on highly activated glyoxyl-agarose, Biomacromolecules 9 (2008) 2553–2561, https://doi.org/10.1021/bm800609g.
[86] V.G. Tacias-Pascacio, S. Peirce, B. Torrestiana-Sanchez, M. Yates, A. Rosales-Quintero, J.J. Virgen-Ortíz, R. Fernandez-Lafuente, Evaluation of different commercial hydrophobic supports for the immobilization of lipases: tuning their stability, activity and specificity, RSC Adv. 6 (2016) 100281–100294, https://doi.org/10.1039/C6RA21730C.
[87] V.G. Tacias-Pascacio, J.J. Virgen-Ortíz, M. Jiménez-Pérez, M. Yates, B. Torrestiana-Sanchez, A. Rosales-Quintero, R. Fernandez-Lafuente, Evaluation of different lipase biocatalysts in the production of biodiesel from used cooking oil: Critical role of the immobilization support, Fuel 200 (2017) 1–10, https://doi.org/10.1016/J.FUEL.2017.03.054.
[88] A. Chowdhury, D. Mitra, D. Biswas, Biolubricant synthesis from waste cooking oil via enzymatic hydrolysis followed by chemical esterification, J. Chem. Technol. Biotechnol. 88 (2013) 139–144, https://doi.org/10.1002/JCTB.3874.
[89] A. Chowdhury, R. Chakraborty, D. Mitra, D. Biswas, Optimization of the production parameters of octyl ester biolubricant using Taguchi's design method and physico-chemical characterization of the product, Ind. Crops Prod. 52 (2014) 783–789, https://doi.org/10.1016/J.INDCROP.2013.11.007.
[90] C. Ortiz, M.L. Ferreira, O. Barbosa, J.C.S. Dos Santos, R.C. Rodrigues, Á. Berenguer-Murcia, L.E. Briand, R. Fernandez-Lafuente, Novozym 435: the "perfect" lipase immobilized biocatalyst? Catal. Sci. Technol. 9 (2019) 2380–2420, https://doi.org/10.1039/c9cy00415g.
[91] R.R. Sousa, M.C.C. Pinto, E.C.G. Aguieiras, E.P. Cipolatti, E.A. Manoel, A.S. da Silva, J.C. Pinto, D.M.G. Freire, V.S. Ferreira-Leitão, Comparative performance and reusability studies of lipases on syntheses of octyl esters with an economic approach, Bioprocess Biosyst. Eng. 45 (2022) 131–145, https://doi.org/10.1007/S00449-021-02646-Z/FIGURES/1.
[92] V. Vescovi, R.L.C. Giordano, A.A. Mendes, P.W. Tardioli, Immobilized lipases on functionalized silica particles as potential biocatalysts for the synthesis of fructose oleate in an organic solvent/water system, Molecules 22 (2017) 212, https://doi.org/10.3390/molecules22020212.
[93] L.F. Ramírez-Verduzco, J.E. Rodríguez-Rodríguez, A.D.R. Jaramillo-Jacob, Predicting cetane number, kinematic viscosity, density and higher heating value of biodiesel from its fatty acid methyl ester composition, Fuel 91 (2012) 102–111, https://doi.org/10.1016/J.FUEL.2011.06.070.
[94] S. Mishra, K.R. Bukkarapu, A. Krishnasamy, A composition based approach to predict density, viscosity and surface tension of biodiesel fuels, Fuel 285 (2021), 119056, https://doi.org/10.1016/J.FUEL.2020.119056.
[95] A.A. Cerón, R.N. Vilas Boas, F.C. Biaggio, H.F. de Castro, Synthesis of biolubricant by transesterification of palm kernel oil with simulated fusel oil: Batch and continuous processes, Biomass Bioenergy 119 (2018) 166–172, https://doi.org/10.1016/J.BIOMBIOE.2018.09.013.
[96] K. Anand, A. Ranjan, P.S. Mehta, Predicting the density of straight and processed vegetable oils from fatty acid composition, Energy Fuels 24 (2010) 3262–3266, https://doi.org/10.1021/EF100143F.
[97] N. Dörmo, K. Bélafi-Bakó, L. Bartha, U. Ehrenstein, L. Gubicza, Manufacture of an environmental-safe biolubricant from fusel oil by enzymatic esterification in solvent-free system, Biochem. Eng. J. 21 (2004) 229–234, https://doi.org/10.1016/j.bej.2004.06.011.
[98] G. Ajithkumar, N.H. Jayadas, M. Bhasi, Analysis of the pour point of coconut oil as a lubricant base stock using differential scanning calorimetry, Lubr. Sci. 21 (2009) 13–26, https://doi.org/10.1002/LS.69.
[99] L. Yao, E.G. Hammond, T. Wang, S. Bhuyan, S. Sundararajan, Synthesis and physical properties of potential biolubricants based on ricinoleic acid, J. Am. Oil Chem. Soc. 87 (2010) 937–945, https://doi.org/10.1007/S11746-010-1574-1.
[100] K.V. Fernandes, A. Papadaki, J.A.C. da Silva, R. Fernandez-Lafuente, A. A. Koutinas, D.M.G. Freire, Enzymatic esterification of palm fatty-acid distillate for the production of polyol esters with biolubricant properties, Ind. Crops Prod. 116 (2018) 90–96, https://doi.org/10.1016/j.indcrop.2018.02.058.
[101] Lexolube HS-C - zslubes. ⟨https://zslubes.com/product/lexolube-hs-c-2/⟩ (Accessed 24October 2021).
[102] Lubricit 10L - zslubes. ⟨https://zslubes.com/product/lubricit-10l-df/⟩ (Accessed 24 October 2021).
[103] Lexolube T-110 - zslubes. ⟨https://zslubes.com/product/lexolube-t-110/⟩ (Accessed 24 October 2021).